

CX 4240 Spring 2026

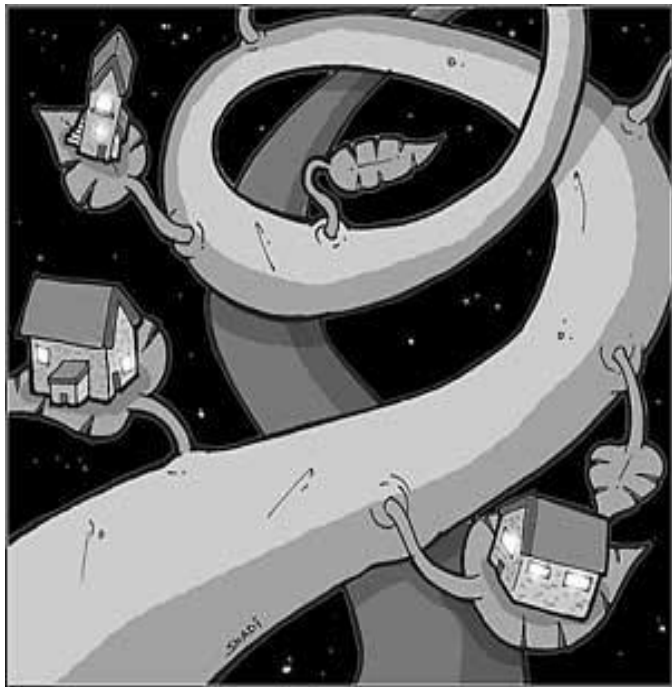
Linear Algebra Revisit

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Basic / Prerequisites

- Probability
 - Distributions, densities, marginalization, conditioning
- Statistics
 - Mean, variance, maximum likelihood estimation
- Linear Algebra and Optimization
 - Vector, matrix, multiplication, inversion, eigen-value decomposition
- Coding Skills
 - Pytorch and/or JAX

Motivational Example: Machine Learning for Apartment Hunting



- Suppose you are to move to Atlanta
- And you want to find the **most reasonably priced** apartment satisfying your **needs**:
monthly rent = $\theta_1(\text{living area}) + \theta_2(\# \text{ bedroom})$

Living area (ft ²)	# bedroom	Monthly rent (\$)
230	1	900
506	2	1800
433	2	1500
190	1	800
...		
150	1	?
270	1.5	?

Linear Regression Model

- Assume y is a linear function of x (features) plus noise ϵ
monthly rent = θ_1 (living area) + θ_2 (# bedroom)
$$y = \theta_0 + \theta_1 x_1 + \cdots + \theta_n x_n + \epsilon$$

where ϵ is an error model as Gaussian $N(0, \sigma^2)$

Probability



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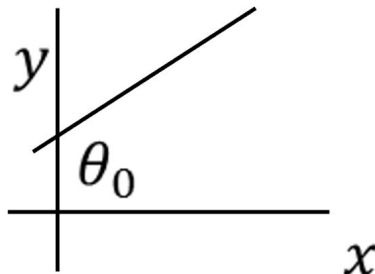
Probability

- Let $\theta = (\theta_0, \theta_1, \dots, \theta_n)^\top$, and augment data by one dimension

Linear algebra $x \leftarrow (1, x)^\top$

Then $y = \theta^\top x + \epsilon$

Linear algebra



Probabilistic Interpretation of Least Mean Square

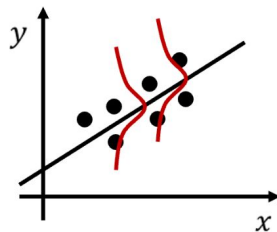
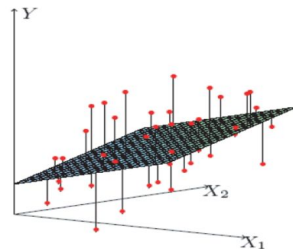
- Assume y is a linear in x plus noise ϵ

$$y = \theta^\top x + \epsilon$$

Linear algebra

- Assume ϵ follows a Gaussian $N(0, \sigma)$

$$p(y^i | x^i; \theta) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(y^i - \theta^\top x^i)^2}{2\sigma^2}\right)$$



Probabilistic Interpretation

- Hence the log-likelihood is:

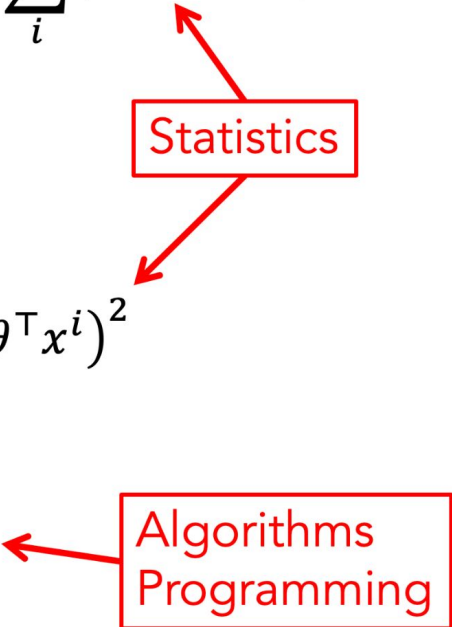
$$\log L(\theta) = m \log \frac{1}{\sqrt{2\pi}\sigma} - \frac{1}{2\sigma^2} \sum_i^m (y^i - \theta^\top x^i)^2$$

- Least Mean Square (LMS)

$$LMS: \frac{1}{m} \sum_i^m (y^i - \theta^\top x^i)^2$$

- How to make it work in real data?

Statistics



Algorithms
Programming

Matrix version of the gradient

- Define $X = (x^1, x^2, \dots, x^m)$, $y = (y^1, y^2, \dots, y^m)^\top$, gradient becomes

Linear algebra $\rightarrow$
$$\frac{\partial L(\theta)}{\partial \theta} = -\frac{2}{m} Xy + \frac{2}{m} XX^\top \theta$$

Linear algebra $\rightarrow$
$$\Rightarrow \hat{\theta} = (XX^\top)^{-1} Xy$$

Algorithms
Programming

- Matrix inversion in $\hat{\theta} = (XX^\top)^{-1} Xy$ **expensive** to compute

- Gradient descent

$$\hat{\theta}^{t+1} \leftarrow \hat{\theta}^t + \frac{\alpha}{m} \sum_i^m (y^i - \hat{\theta}^{t^\top} x^i) x^i$$

Optimization

Usage in Modern ML

- Model Design
 - Convolution Operation
 - Attention Design in Transformer
 - etc
- PyTorch or JAX Implementation
 - Matrix Ops for Acceleration

Revisit of Linear Algebra

- Basics
- Dot and Vector Products
- Identity, Diagonal and Orthogonal Matrices
- Trace
- Norms
- Inverse of a matrix
- Eigenvalues and Eigenvectors
- Singular Value Decomposition
- Matrix Calculus (Optional)

Linear Algebra Basics - I

- Linear algebra provides a way of compactly representing and operating on sets of linear equations

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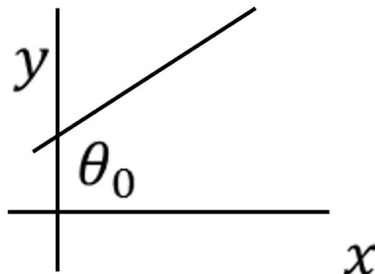
Probability

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Linear algebra



Linear Algebra Basics - I

- Linear algebra provides a way of compactly representing and operating on sets of linear equations

$$4x_1 - 5x_2 = -13 \qquad -2x_1 + 3x_2 = 9$$

can be written in the form of

$$A = \begin{bmatrix} 4 & 5 \\ -2 & 3 \end{bmatrix} \quad b = \begin{bmatrix} -13 \\ 9 \end{bmatrix} \quad Ax = b$$

- $A \in \mathbb{R}^{m \times n}$ denotes a matrix with m rows and n columns, where elements belong to real numbers.
- $x \in \mathbb{R}^n$ denotes a vector with n real entries. By convention an n dimensional vector is often thought as a matrix with n rows and 1 column.

Linear Algebra Basics - II

- Transpose of a matrix results from flipping the rows and columns. Given $A \in \mathbb{R}^{m \times n}$, transpose is $A^T \in \mathbb{R}^{n \times m}$
- For each element of the matrix, the transpose can be written as $A^T_{ij} = A_{ji}$

A

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}$$

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- The following properties of the transposes are easily verified

$$(A^T)^T = A$$

$$(AB)^T = B^T A^T$$

$$(A + B)^T = A^T + B^T$$

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- A square matrix $A \in \mathbb{R}^{n \times n}$ is symmetric if $A = A^T$ and it is anti-symmetric if $A = -A^T$. Thus each matrix can be written as a sum of symmetric and anti-symmetric matrices.

$$C = \frac{1}{2}(C + C^T) + \frac{1}{2}(C - C^T)$$

Vector and Matrix Multiplication - I

- The product of two matrices $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times p}$ is given by $C \in \mathbb{R}^{m \times p}$, where $C_{ij} = \sum_{k=1}^n A_{ik} B_{kj}$

Diagram illustrating the dot product calculation for matrix multiplication:

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \times \begin{bmatrix} 7 & 8 \\ 9 & 10 \\ 11 & 12 \end{bmatrix} = \begin{bmatrix} 58 & \dots \\ \dots & \dots \end{bmatrix}$$

The calculation shown is the dot product of the first row of the first matrix (1, 2, 3) and the first column of the second matrix (7, 9, 11), resulting in 58.

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The diagram shows the multiplication of two matrices, A and B, to produce matrix C. Matrix A is a 2x3 matrix with elements 1, 2, 3 in the first row and 4, 5, 6 in the second row. Matrix B is a 3x2 matrix with elements 7, 8 in the first column and 9, 10 in the second column, and 11, 12 in the third column. The result matrix C is a 2x2 matrix with elements 58 and 64 in the first row. The elements 1, 2, 3, 8, and 64 are highlighted in yellow. A blue 'x' symbol is between the matrices, and a blue '=' symbol is between the matrices and the result. A yellow curved arrow points from the first row of A to the first row of C, and another yellow curved arrow points from the second column of B to the second column of C.

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \times \begin{bmatrix} 7 & 8 \\ 9 & 10 \\ 11 & 12 \end{bmatrix} = \begin{bmatrix} 58 & 64 \end{bmatrix}$$

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$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \times \begin{bmatrix} 7 & 8 \\ 9 & 10 \\ 11 & 12 \end{bmatrix} = \begin{bmatrix} 58 & 64 \\ 139 & 154 \end{bmatrix} \checkmark$$

Vector and Matrix Multiplication - I

- Given two vectors $x, y \in \mathbb{R}^n$, the term $x^\top y$ (also $x \cdot y$) is called the *inner product* or *dot product* of the vectors, and is a real number given by $\sum_{i=1}^n x_i y_i$. For example,

$$x^\top y = [x_1 \quad x_2 \quad x_3] \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \sum_{i=1}^3 x_i y_i$$

Vector and Matrix Multiplication - III

- Given two vectors $x \in \mathbb{R}^n, y \in \mathbb{R}^m$, the term xy^T is called the **outer product** of the vectors, and is a matrix given by $(xy^T)_{ij} = x_i y_j$. For example,

$$xy^T = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \begin{bmatrix} y_1 & y_2 & y_3 \end{bmatrix} = \begin{bmatrix} x_1 y_1 & x_1 y_2 & x_1 y_3 \\ x_2 y_1 & x_2 y_2 & x_2 y_3 \\ x_3 y_1 & x_3 y_2 & x_3 y_3 \end{bmatrix}$$

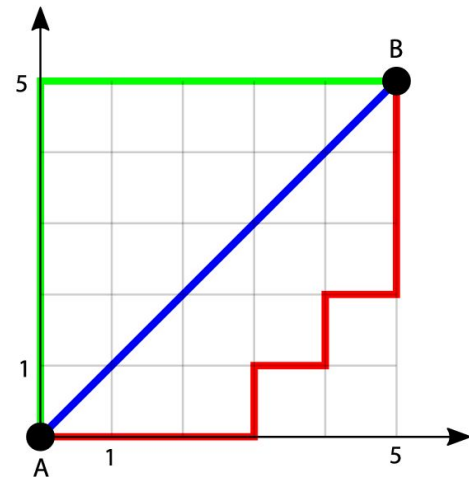
Norms - I

- Norm of a vector $\|x\|$ is informally a measure of the "length" of a vector
- More formally, a norm is any function $f: \mathbb{R}^n \rightarrow \mathbb{R}$ that satisfies
 - For all $x \in \mathbb{R}^n$, $f(x) \geq 0$ (non-negativity)
 - $f(x) = 0$ if and only if $x = 0$ (definiteness)
 - For $x \in \mathbb{R}^n$, $t \in \mathbb{R}$, $f(tx) = |t|f(x)$ (homogeneity)
 - For all $x, y \in \mathbb{R}^n$, $f(x + y) \leq f(x) + f(y)$ (triangle inequality)

Norms - III

- Common norms used in machine learning are

- ℓ_2 norm: $\|x\|_2 = \sqrt{\sum_{i=1}^n x_i^2}$
- ℓ_1 norm: $\|x\|_1 = \sum_{i=1}^n |x_i|$
- ℓ_∞ norm: $\|x\|_\infty = \max_i |x_i|$



- All norms presented so far are examples of the family of ℓ_p norms, which are parameterized by a real number $p \geq 1$:

$$\|x\|_p = (\sum_{i=1}^n |x_i|^p)^{\frac{1}{p}}$$

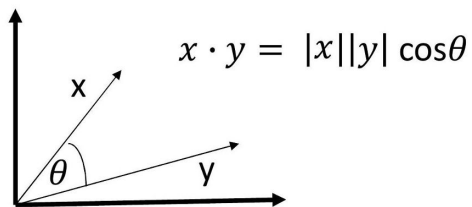
- Norms can be defined for matrices, such as the Frobenius norm.

$$\|A\|_F = \sqrt{\sum_{i=1}^m \sum_{j=1}^n A_{ij}^2}$$

Norm Revisit

$$x^T y = [x_1 \quad x_2 \quad x_3] \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \sum_{i=1}^3 x_i y_i$$

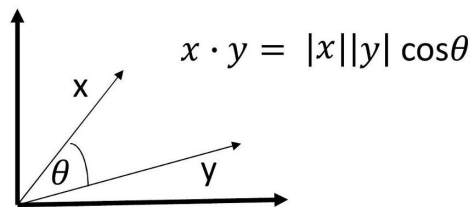
- The dot product also has a geometrical interpretation, for vectors in $x, y \in \mathbb{R}^2$ with angle θ between them



Norm Revisit

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- The dot product also has a geometrical interpretation, for vectors in $x, y \in \mathbb{R}^2$ with angle θ between them



which leads to use of dot product for testing orthogonality, getting the Euclidean norm of a vector, and scalar projections.

Trace of a Matrix

- The trace of a matrix $A \in \mathbb{R}^{n \times n}$, denoted as $\mathbf{tr}(\mathbf{A})$, is the sum of the diagonal elements in the matrix

$$\mathrm{tr}(A) = \sum_{i=1}^n A_{ii}$$

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- The trace has the following properties
 - For $A \in \mathbb{R}^{n \times n}$, $\mathbf{tr}(A) = \mathbf{tr}A^\top$
 - For $A, B \in \mathbb{R}^{n \times n}$, $\mathbf{tr}(A + B) = \mathbf{tr}(A) + \mathbf{tr}(B)$
 - For $A \in \mathbb{R}^{n \times n}$, $t \in \mathbb{R}$, $\mathbf{tr}(tA) = t \cdot \mathbf{tr}(A)$
 - For A, B, C such that ABC is a square matrix $\mathbf{tr}(ABC) = \mathbf{tr}(BCA) = \mathbf{tr}(CAB)$

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 - For $A \in \mathbb{R}^{n \times n}$, $t \in \mathbb{R}$, $\mathbf{tr}(tA) = t \cdot \mathbf{tr}(A)$
 - For A, B, C such that ABC is a square matrix $\mathbf{tr}(ABC) = \mathbf{tr}(BCA) = \mathbf{tr}(CAB)$
- The trace of a matrix helps us easily compute norms and eigenvalues of matrices as we will see later

$$\|A\|_F = \sqrt{\sum_{i=1}^m \sum_{j=1}^n A_{ij}^2} = \sqrt{\mathbf{tr}(A^\top A)}$$

Identity Matrices

- The identity matrix, denoted by $I \in \mathbb{R}^{n \times n}$ is a square matrix with ones on the diagonal and zeros everywhere else

$$I_1 = [1],$$

$$I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

$$I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

.....,

$$I_n = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ .. & .. & .. & .. & .. & .. \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

Diagonal Matrices

- A diagonal matrix is matrix where all non-diagonal matrices are 0 . This is typically denoted as $D = \text{diag}(d_1, d_2, d_3, \dots, d_n)$

$$D = \begin{bmatrix} d_1 & 0 & \dots & 0 \\ 0 & d_2 & 0 & 0 \\ \vdots & 0 & \ddots & \vdots \\ 0 & 0 & \dots & d_n \end{bmatrix}$$

Orthogonal Matrices

- Two vectors $x, y \in \mathbb{R}^n$ are orthogonal if $x^\top y = 0$. A square matrix $U \in \mathbb{R}^{n \times n}$ is orthogonal if all its columns are orthogonal to each other and are normalized
- It follows from orthogonality and normality that
 - $U^\top U = I = UU^\top$
 - $\|Ux\|_2 = \|x\|_2$

Inverse of a Matrix

- The inverse of a square matrix $A \in \mathbb{R}^{n \times n}$ is denoted A^{-1} and is the unique matrix such that $A^{-1}A = I = AA^{-1}$

$$\begin{bmatrix} 3 & 4 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ -\frac{1}{2} & \frac{3}{2} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Education Portal

Inverse of a Matrix

- The inverse of a square matrix $A \in \mathbb{R}^{n \times n}$ is denoted A^{-1} and is the unique matrix such that $A^{-1}A = I = AA^{-1}$
- For some square matrices A^{-1} may not exist, and we say that A is *singular or non-invertible*. In order for A to have an inverse, A must be *full rank*.
- For non-square matrices the inverse, denoted by A^+ , is given by $A^+ = (A^T A)^{-1} A^T$ called the *pseudo inverse*

Determinant and Inverse of a Matrix

- The determinant of a square matrix $A \in \mathbb{R}^{n \times n}$ is a function $f : \mathbb{R}^{n \times n} \rightarrow \mathbb{R}$, denoted by $|A|$ or $\det A$, and is calculated as

$$|A| = \sum_{i=1}^n (-1)^{i+j} a_{ij} |A_{\setminus i, \setminus j}| \quad (\text{for any } j \in 1, 2, \dots, n)$$

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Linear Independence and Rank

- A set of vectors $\{x_1, x_2, \dots, x_n\} \subset \mathbb{R}^m$ are said to be *(linearly) independent* if no vector can be represented as a linear combination of the remaining vectors. That is if

$$x_n = \sum_{i=1}^{n-1} \alpha_i x_i$$

- for some scalar values $\alpha_1, \alpha_2, \dots \in \mathbb{R}$ then we say that the vectors are linearly *dependent*; otherwise the vectors are linearly independent
- The *column rank* of a matrix $A \in \mathbb{R}^{m \times n}$ is the size of the largest subset of columns of A that constitute a linearly independent set. *Row rank* of a matrix is defined similarly for rows of a matrix.

Range and Null Space

- The span of a set of vectors $\{x_1, x_2, \dots, x_n\}$ is the set of all vectors that can be expressed as a linear combination of the set $\{v: v = \sum_{i=1}^n \alpha_i x_i, \alpha_i \in \mathbb{R}\}$
- If $\{x_1, x_2, \dots, x_n\} \in \mathbb{R}^n$ is a set of linearly independent set of vectors, then $\text{span}(\{x_1, x_2, \dots, x_n\}) = \mathbb{R}^n$
- The range of a matrix $A \in \mathbb{R}^{m \times n}$, denoted as $\mathcal{R}(A)$, is the span of the columns of A
- The nullspace of a matrix $A \in \mathbb{R}^{m \times n}$, denoted $\mathcal{N}(A)$, is the set of all vectors that equal 0 when multiplied by A
 - $\mathcal{N}(A) = \{x \in \mathbb{R}^n: Ax = 0\}$

Column and Row Space

- The row space and column space are the linear subspaces generated by row and column vectors of a matrix
- Linear subspace, is a vector space that is a subset of some other higher dimension vector space
- For a matrix $A \in \mathbb{R}^{m \times n}$
 - $Col\ space(A) = span(columns\ of\ A)$
 - $Rank(A) = dim(rowspace(A)) = dim(colspace(A))$

Eigenvalues and Eigenvectors - I

- Given a square matrix $A \in \mathbb{R}^{n \times n}$ we say that $\lambda \in \mathbb{C}$ is an eigenvalue of A and $x \in \mathbb{C}^n$ is an eigenvector if

$$Ax = \lambda x, x \neq 0$$

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$$Ax = \lambda x, x \neq 0$$

- Intuitively this means that upon multiplying the matrix A with a vector x , we get the same vector, but scaled by a parameter λ
- Geometrically, we are transforming the matrix A from its original orthonormal basis/co-ordinates to a new set of orthonormal basis x with magnitude as λ

Eigenvalues and Eigenvectors - II

- All the eigenvectors can be written together as $AX = X\Lambda$ where the diagonals of X are the eigenvectors of A , and Λ is a diagonal matrix whose elements are eigenvalues of A

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- All the eigenvectors can be written together as $AX = X\Lambda$ where the columns of X are the eigenvectors of A , and Λ is a diagonal matrix whose elements are eigenvalues of A
- If the eigenvectors of A are invertible, then $A = X\Lambda X^{-1}$

Eigenvalues and Eigenvectors - II

- All the eigenvectors can be written together as $AX = X\Lambda$ where the diagonals of X are the eigenvectors of A , and Λ is a diagonal matrix whose elements are eigenvalues of A
- If the eigenvectors of A are invertible, then $A = X\Lambda X^{-1}$
- There are several properties of eigenvalues and eigenvectors
 - $\text{Tr}(A) = \sum_{i=1}^n \lambda_i$
 - $|A| = \prod_{i=1}^n \lambda_i$
 - Rank of A is the number of non-zero eigenvalues of A
 - If A is non-singular then $\frac{1}{\lambda_i}$ are the eigenvalues of A^{-1}
 - The eigenvalues of a diagonal matrix are the diagonal elements of the matrix itself!

Eigenvalues and Eigenvectors - III

- For a symmetric matrix A , it can be shown that eigenvalues are real and the eigenvectors are orthonormal. Thus it can be represented as $U\Lambda U^T$
- Considering quadratic form of A ,

$$x^T A x = x^T U \Lambda U^T x = y^T \Lambda y = \sum_{i=1}^n \lambda_i y_i^2 \quad (\text{where } y = U^T x)$$

- Since y_i^2 is always positive the sign of the expression always depends on λ_i . If $\lambda_i > 0$ then the matrix A is positive definite, if $\lambda_i \geq 0$ then the matrix A is positive semidefinite
- For a multivariate Gaussian, the variances of x and y do not fully describe the distribution. The eigenvectors of this covariance matrix capture the directions of highest variance and eigenvalues the variance

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Eigenvalues and Eigenvectors - IV

- We can rewrite the original equation in the following manner

$$\begin{aligned}Ax &= \lambda x, x \neq 0 \\ \Rightarrow (\lambda I - A)x &= 0, x \neq 0\end{aligned}$$

- This is only possible if $(\lambda I - A)$ is singular, that is $|\lambda I - A| = 0$.
- Thus, eigenvalues and eigenvectors can be computed.
 - Compute the determinant of $A - \lambda I$.
 - This results in a polynomial of degree n .
 - Find the roots of the polynomial by equating it to zero.
 - The n roots are the n eigenvalues of A . They make $A - \lambda I$ singular.
 - For each eigenvalue λ , solve $(A - \lambda I)x = 0$ to find an eigenvector x

Singular Value Decomposition

- Singular value decomposition, known as SVD, is a factorization of a real matrix with applications in calculating pseudo-inverse, rank, solving linear equations, and many others.
- For a matrix $M \in \mathbb{R}^{m \times n}$ assume $n \leq m$
 - $M = U\Sigma V^T$ where $U \in \mathbb{R}^{m \times m}, V^T \in \mathbb{R}^{n \times n}, \Sigma \in \mathbb{R}^{m \times n}$
 - The m columns of U , and the n columns of V are called the left and right singular vectors of M . The diagonal elements of Σ, Σ_{ii} are known as the singular values of M .
 - Let v be the i^{th} column of V , and u be the i^{th} column of U , and σ be the i^{th} diagonal element of Σ

$$Mv = \sigma u \text{ and } M^T u = \sigma v$$

Singular Value Decomposition - II

$$M = [u_1 \ u_2 \ \dots \ u_n] \begin{bmatrix} \Sigma_{11} & \dots & \Sigma_{1n} \\ \vdots & \ddots & \vdots \\ \Sigma_{m1} & \dots & \Sigma_{mn} \end{bmatrix} [v_1 \ v_2 \ \dots \ v_n]^T$$

Diagram illustrating the Singular Value Decomposition (SVD) of matrix M :

- principal directions**: $[u_1 \ u_2 \ \dots \ u_n]$
- Scaling factor**: $\begin{bmatrix} \Sigma_{11} & \dots & \Sigma_{1n} \\ \vdots & \ddots & \vdots \\ \Sigma_{m1} & \dots & \Sigma_{mn} \end{bmatrix}$
- Projection in principal directions**: $[v_1 \ v_2 \ \dots \ v_n]^T$

- Singular value decomposition is related to eigenvalue decomposition
 - Then covariance matrix is $C = \frac{1}{m}XX^T$
 - Starting from singular vector pair
 - $M^T u = \sigma v$
 - $\Rightarrow MM^T u = \sigma Mv$
 - $\Rightarrow MM^T u = \sigma^2 u$
 - $\Rightarrow Cu = \lambda u$

Matrix Calculus

- For a vector $x, b \in \mathbb{R}^n$, let $f(x) = b^\top x$, then $\nabla_x b^\top x$ is equal to b
 - $\frac{\partial f(x)}{\partial x_k} = \frac{\partial}{\partial x_k} \sum_{i=1}^n b_i x_i = b_k$
- Now for a quadratic function, $f(x) = x^\top A x$, with $A \in \mathbb{S}^n$, $\frac{\partial f(x)}{\partial x_k} = 2Ax$
 - $$\begin{aligned}\frac{\partial f(x)}{\partial x_k} &= \frac{\partial}{\partial x_k} \sum_{i=1}^n \sum_{j=1}^n A_{ij} x_i x_j \\ &= \sum_{i \neq k} A_{ik} x_i + \sum_{j \neq k} A_{kj} x_j + 2A_{kk} x_k \\ &= 2 \sum_{i=1}^n A_{ki} x_i\end{aligned}$$
- Let $f(X) = X^{-1}$, then $\partial(X^{-1}) = -X^{-1}(\partial X)X^{-1}$

References for self study

Resources for review of material

- [Linear Algebra Review and Reference by Zico Kotler](#)
- [Matrix Cookbook by KB Peterson](#)

Back to Apartment Hunting

- Given m data points, find θ that minimizes the mean square error

$$\hat{\theta} = \underset{\theta}{\operatorname{argmin}} L(\theta) = \frac{1}{m} \sum_i^m (y^i - \theta^\top x^i)^2$$

Optimization

Statistics

- Set gradient to 0 and find parameter

Optimization

$$\frac{\partial L(\theta)}{\partial \theta} = -\frac{2}{m} \sum_i^m (y^i - \theta^\top x^i) x^i = 0$$

Linear
algebra

$$\Leftrightarrow -\frac{2}{m} \sum_i^m y^i x^i + \frac{2}{m} \sum_i^m x^i x^{i\top} \theta = 0$$

Statistics

Statistics

Optimization for LMS

- Define $X = (x^1, x^2, \dots, x^m)$, $y = (y^1, y^2, \dots, y^m)^\top$, gradient becomes

Linear algebra $\rightarrow$
$$\frac{\partial L(\theta)}{\partial \theta} = -\frac{2}{m} Xy + \frac{2}{m} XX^\top \theta$$

Linear algebra $\rightarrow$
$$\Rightarrow \hat{\theta} = (XX^\top)^{-1} Xy$$

Algorithms
Programming

- Matrix inversion in $\hat{\theta} = (XX^\top)^{-1} Xy$ expensive to compute

- Gradient descent

$$\hat{\theta}^{t+1} \leftarrow \hat{\theta}^t + \frac{\alpha}{m} \sum_i^m (y^i - \hat{\theta}^{t^\top} x^i) x^i$$

Optimization

Registration

- Friday is the registration deadline.
- If you decide to drop the course, please do so ASAP so that other people on the waitlist have time to register!
- See you next week!

Q&A